Best-prediction ECS — the trust-gated, conservation-gated estimate (widened)

Status: computed and guarded — src/pacland/climate/ecs_model.py + tests/test_climate_ecs_model.py (12 tests, green). The first output of "the model that works": not a certified future (impossible), but the best possible ECS estimate built from the reliable content only.

The finding, in one sentence

> **Two views of the same gates bracket the equilibrium climate sensitivity at ~2.7–3.1 K — a widened 53-model ensemble gated on sensitivity + forcing lands at 3.08 K (15 models, 13 families), while the strictest three-axis gate (which adds conservation, where data exists) collapses to 2.71 K (one family) — both inside the IPCC likely range, at its best estimate (~3.0 K).**

The method — hard gates, not arbitrary weights

The future cannot be validated, so the best prediction is the one built only on the models that pass the checks we can actually run. Each axis is a **hard pass/fail gate** (parameter-free — no arbitrary continuous weights an adversarial reviewer could attack):

gate	what it requires	source
conservation	closes energy, mass AND salt in its control run	Irving et al. 2021 (ours)
sensitivity	ECS inside the IPCC likely range 2.5–4.0 K	Zelinka et al. 2020
forcing	ERF2x inside 3.3–4.1 W/m²	Zelinka et al. 2020

Conservation drift exists for only 19 of the models (Irving's subset), so the two views are reported over their own covers and never mixed:

view	cover	gates	result
widened	53 Zelinka models	sensitivity + forcing	15 models, 13 families
strict	19 models with drift	all three axes	2 models, 1 family

The numbers (`python3 -m pacland.climate.ecs_model`)

```
WIDENED (53 models, sensitivity + forcing gates):
  naive (53)       3.78 K  [1.83, 5.64]
  sens+forc (15)   3.08 K  [2.56, 3.87]   (13 families)
STRICT (19 models with conservation drift, all three axes):
  naive (19)       4.08 K  [2.71, 5.55]
  conservation ( 5) 3.86 K
  three-axis ( 2)    2.71 K  (1 family)
```

The two results, and what each is for

Widened — 3.08 K (15 models, 13 families). The full 53-model ensemble, gated on the two axes that exist for every model. It removes 0.70 K of the naive mean and lands at the IPCC best estimate with a *real multi-family

ensemble* (AWI, CMCC, EC-Earth, FGOALS, GFDL, GISS, MIROC, MPI, MRI, NorCPM, NorESM, SAM0 …). This is the best prediction: diverse, defensible, parameter-free.

Strict — 2.71 K (2 GISS variants, one family). Adding the conservation gate (where the data exists) removes 1.37 K of the hot-model bias and collapses to the single family that does not leak physics, is not hot, and forces correctly. This is the strictest subset, not an ensemble — but it *agrees* with the widened answer to within the IPCC range.

The bracket [2.71, 3.08] K spans the IPCC best estimate (3.0 K). Two independent ways of applying the same discipline land on the same answer, which is exactly the Rule of Three: the widened gate (many families) and the strict gate (conservation) bracket the field's own best estimate, achieved by verification rather than averaging.

The honest caveats (each made a test)

1. The 19-model drift subset is hot-biased — its naive mean (4.08 K) is higher than the full 53's (3.78 K), so the two naives differ and are never mixed.
2. The strict gate collapses to one family (the GISS pair) — conservation is so selective that only one family survives; the widened gate (13 families) is the ensemble result, the strict gate is the cross-check.
3. Family is a naming heuristic (leading non-digit prefix) — good enough to show "many families vs one", not a taxonomy.

Why this helps the sector

- **A defensible best prediction with zero tunable parameters** — anyone can run the same gates over their own model set and get the same number.
- **It separates two questions the field conflates**: *how hot* (ECS, gated 2.7–3.1 K) vs *which models to trust* (the 15 or the 2 that pass).
- **It names the direction**: to sharpen the estimate, add models that *pass the gates* (conserve, reasonable sensitivity, reasonable forcing) — not another point forecast.

Honest limits

- **Not a new climate model** — this re-weights the existing ensemble with verification gates; it simulates nothing.
- **Thresholds are assessed ranges, not measurements** — the *ranking* is robust to them, the exact survivor count is stated at these thresholds (same honesty rule as docs/climate-cmip6-drift.md).
- **Conservation data covers 19 of 53 models** — the three-axis gate is only meaningful on that subset; the widened gate is the general result.

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.ecs_model
PYTHONPATH=src python3 -m unittest tests.test_climate_ecs_model -v   # 12 tests
```

Data: data/climate/cmip6_ecs.json (full 53-model Zelinka et al. 2020 table)
× data/climate/cmip6_drift.json (19-model Irving et al. 2021 drift). The
three-axis verdict is src/pacland/climate/sensitivity_crosscheck.py.